



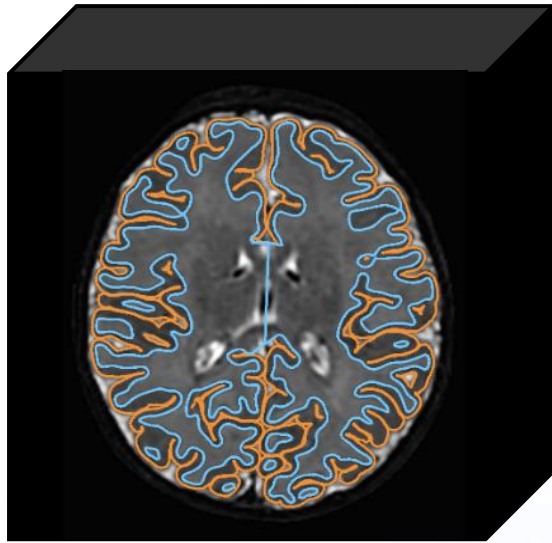
IEEE
ISBI 2026

2026 IEEE International Symposium on Biomedical Imaging
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NC-Reg : Neural Cortical Maps for Rigid Registration

***Ines Vati**^{1,2}, Pierrick Bourgeat¹, Rodrigo Santa Cruz², Vincent Dore¹, Olivier Salvado², Clinton Fookes², Léo Lebrat²*

Background

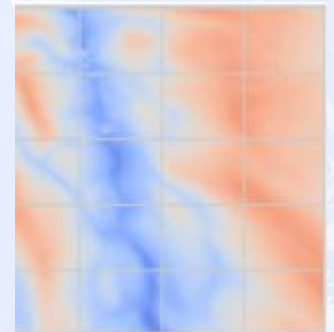
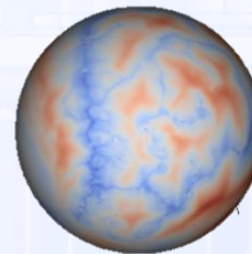
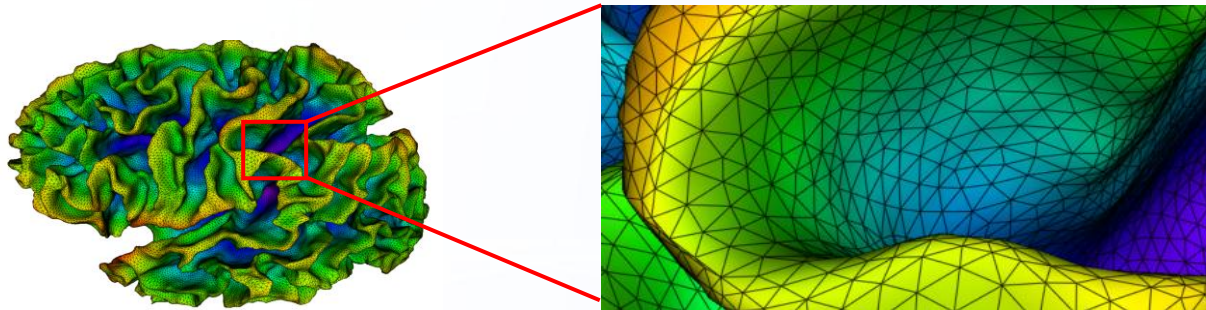


MRI

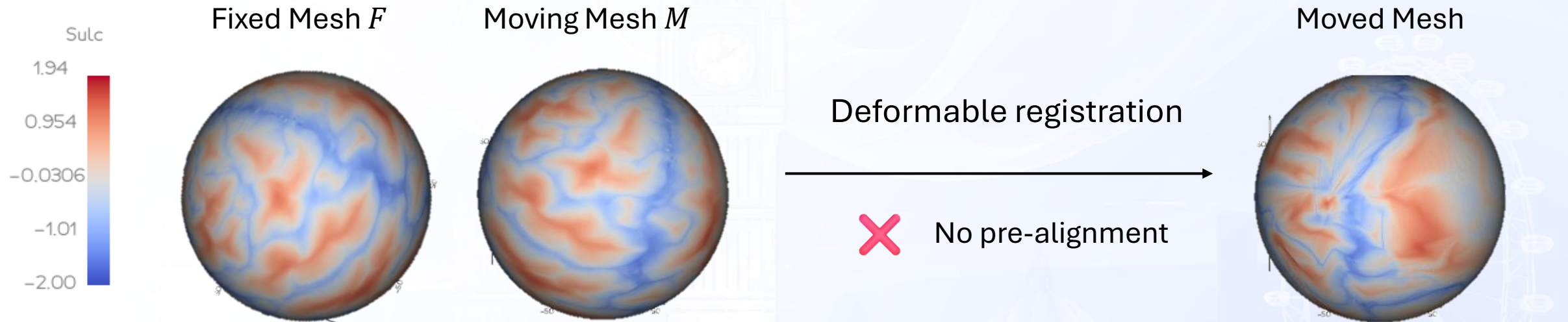


Projection for
faster
interpolation

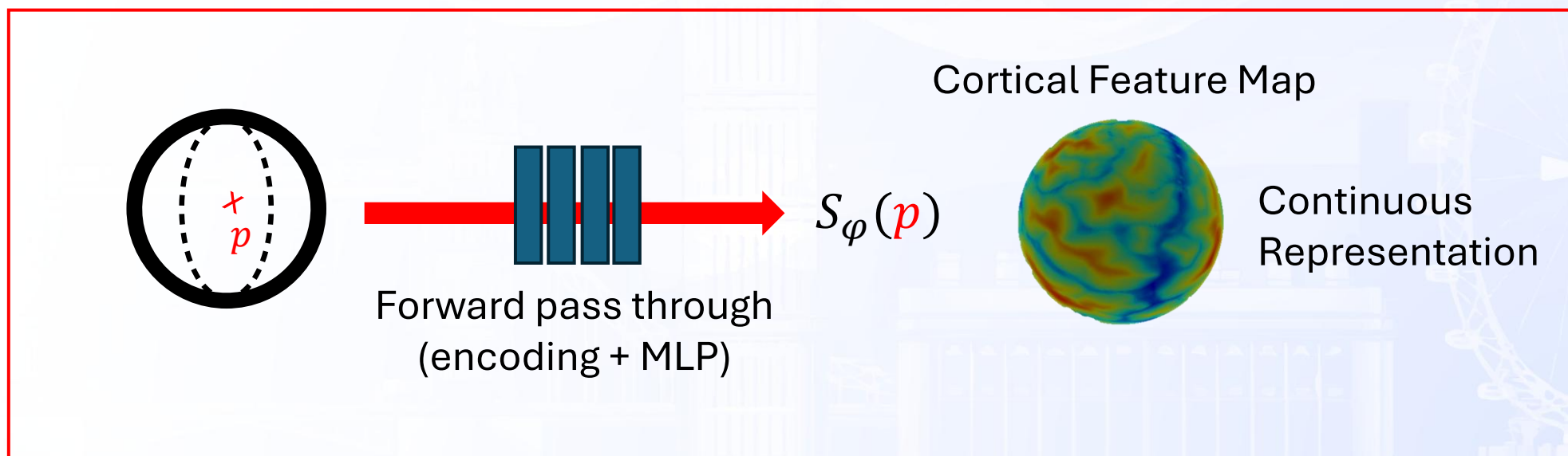
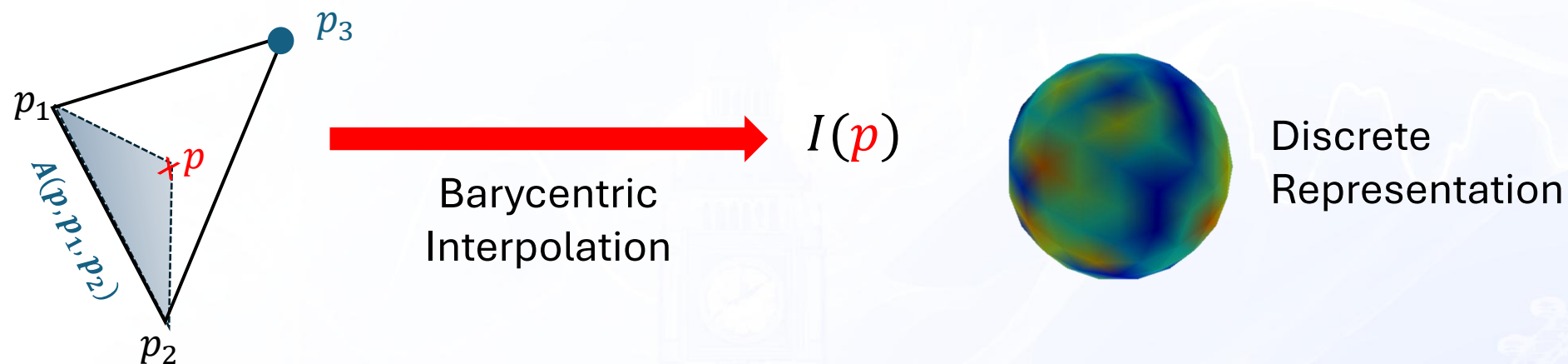
Unstructured **discrete** data



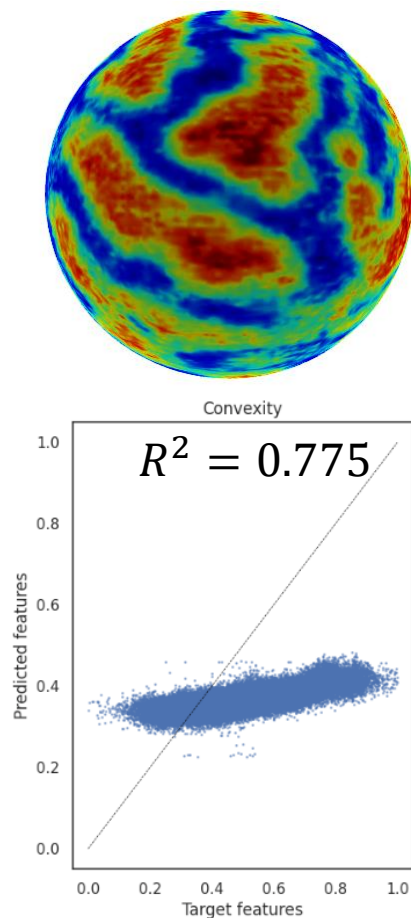
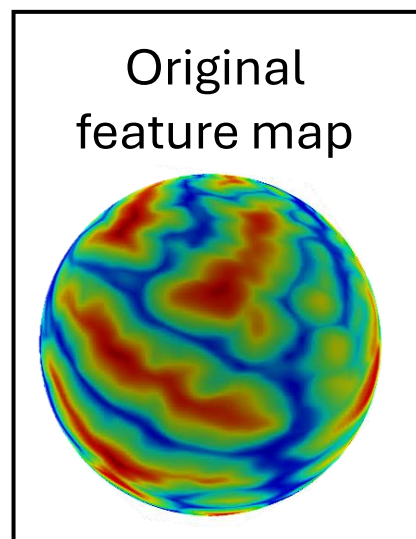
Background



Neural Cortical (NC) Maps

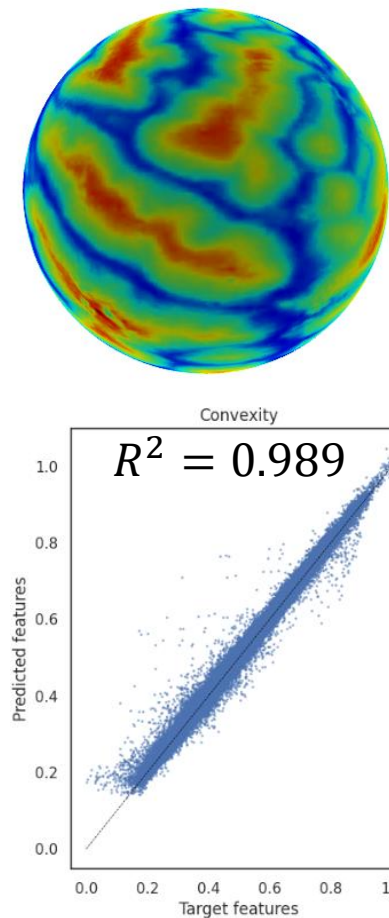


Neural Cortical (NC) Maps



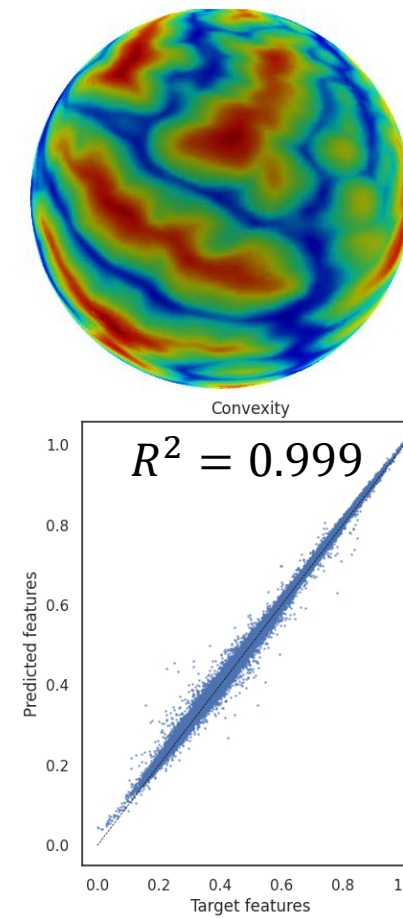
$N_{\text{iter}} = 10$

0.7s



$N_{\text{iter}} = 100$

0.8s



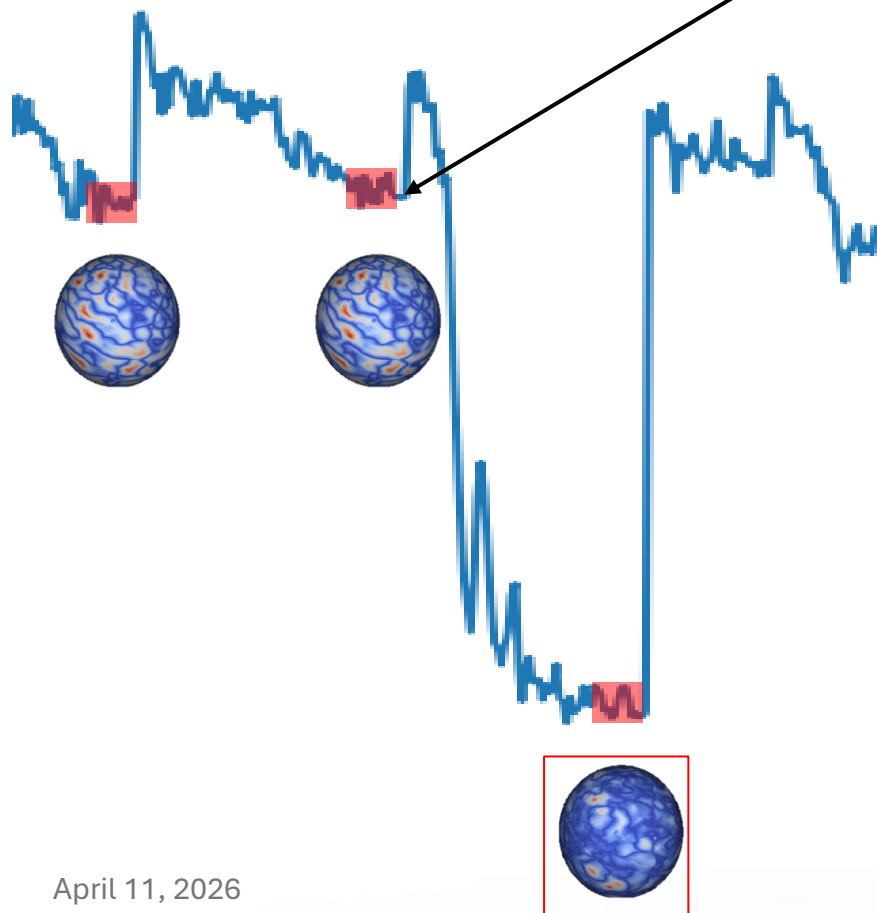
$N_{\text{iter}} = 1000$

6.6s

Time

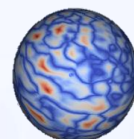
NC-Reg with Simulated Annealing (SA)

$$\Delta\mathcal{L} < 0 \quad \text{OR} \quad \mathcal{U}([0,1]) < \exp\left(\frac{-\Delta\mathcal{L}}{T_t}\right)$$



Local minimum

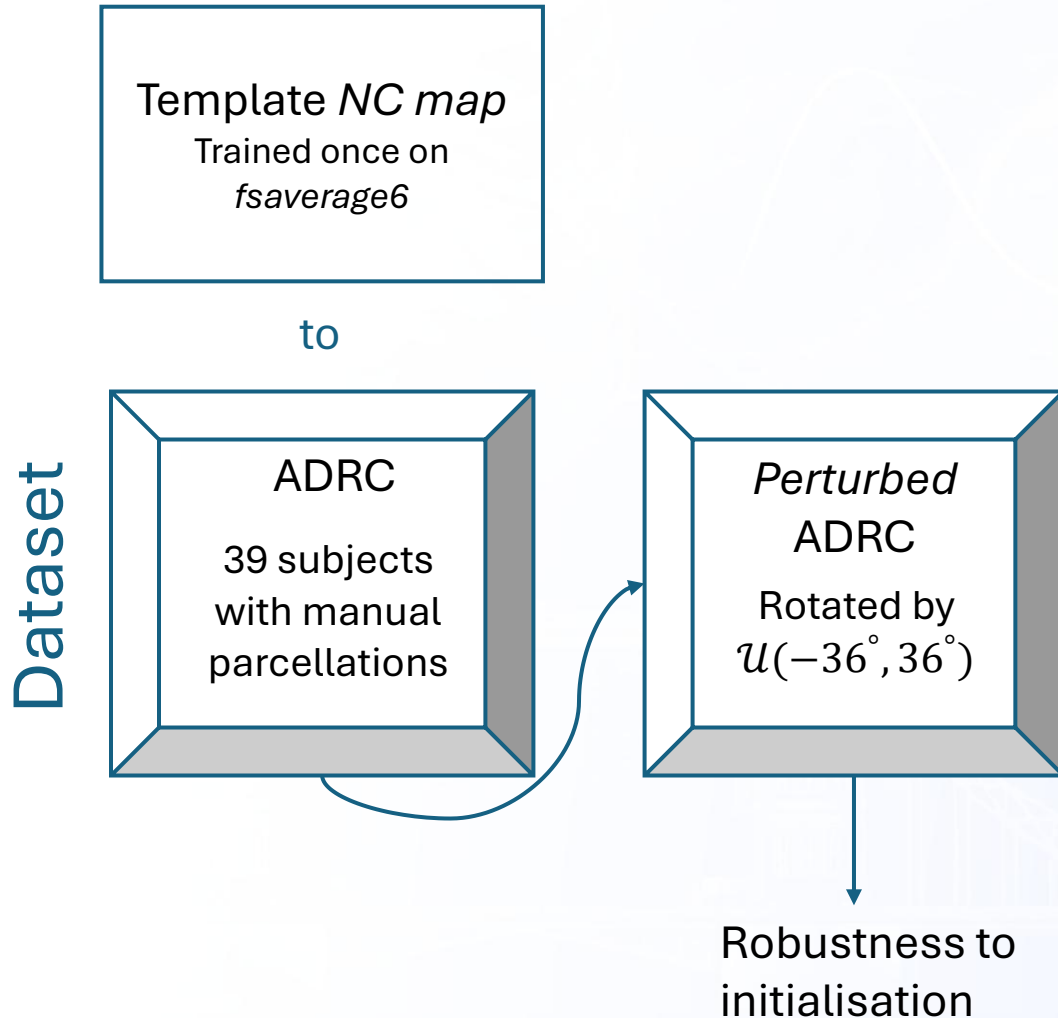
$$\Delta\mathcal{L} = \mathcal{L}(F, M, R_{rand}) - \mathcal{L}_t$$



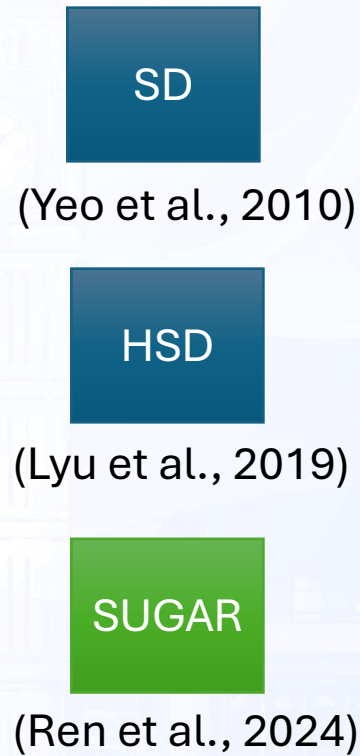
Feature difference

Experimental Setup

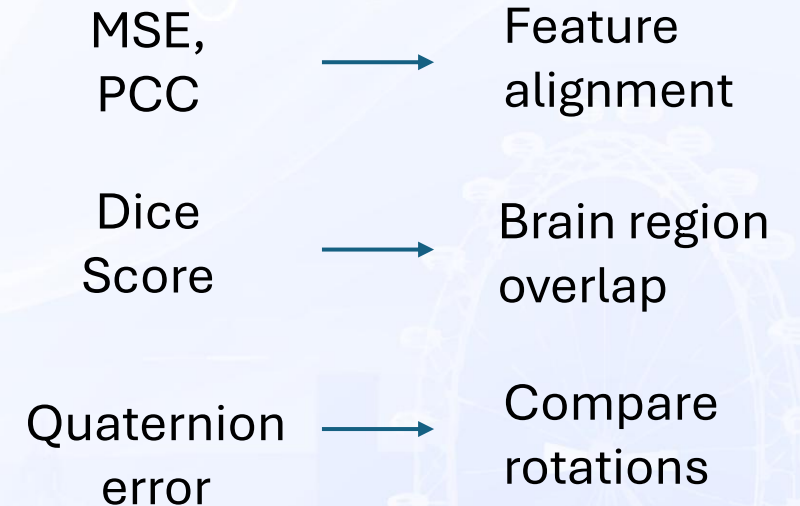
Task: Register



Baselines

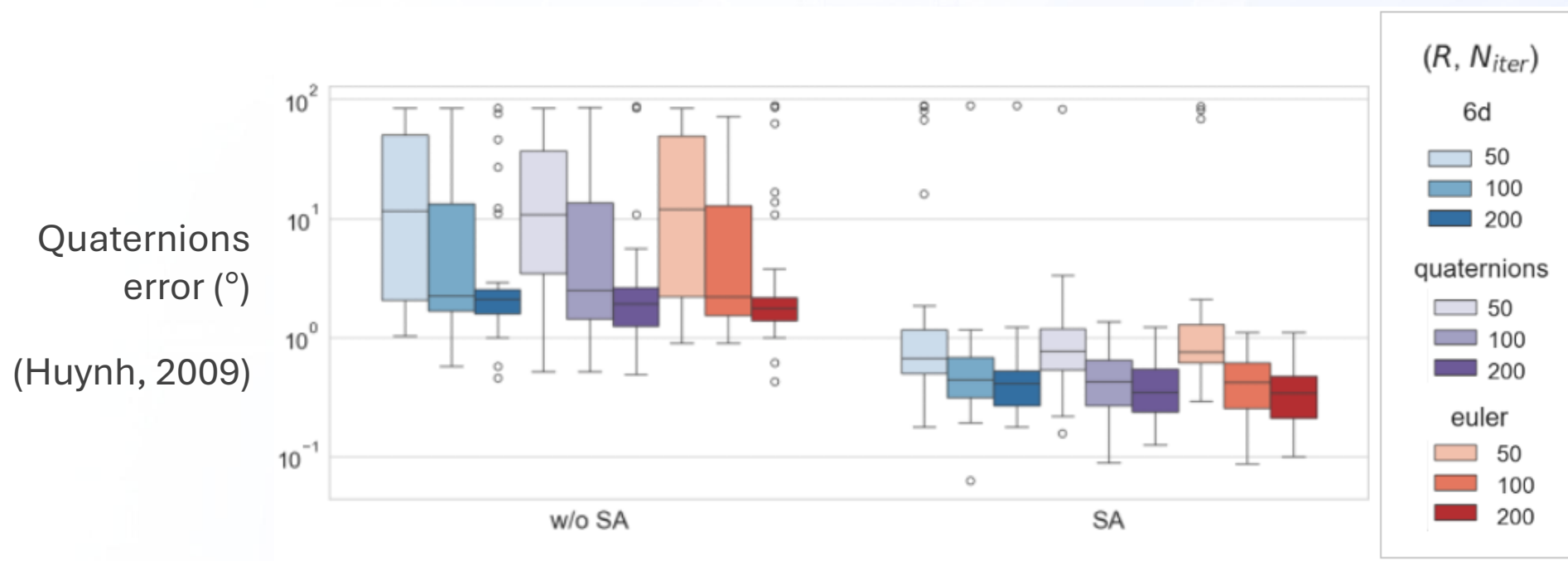


Evaluation metrics



NC-Reg : Relevance of SA

Same **solution** for **different** initialisation ?

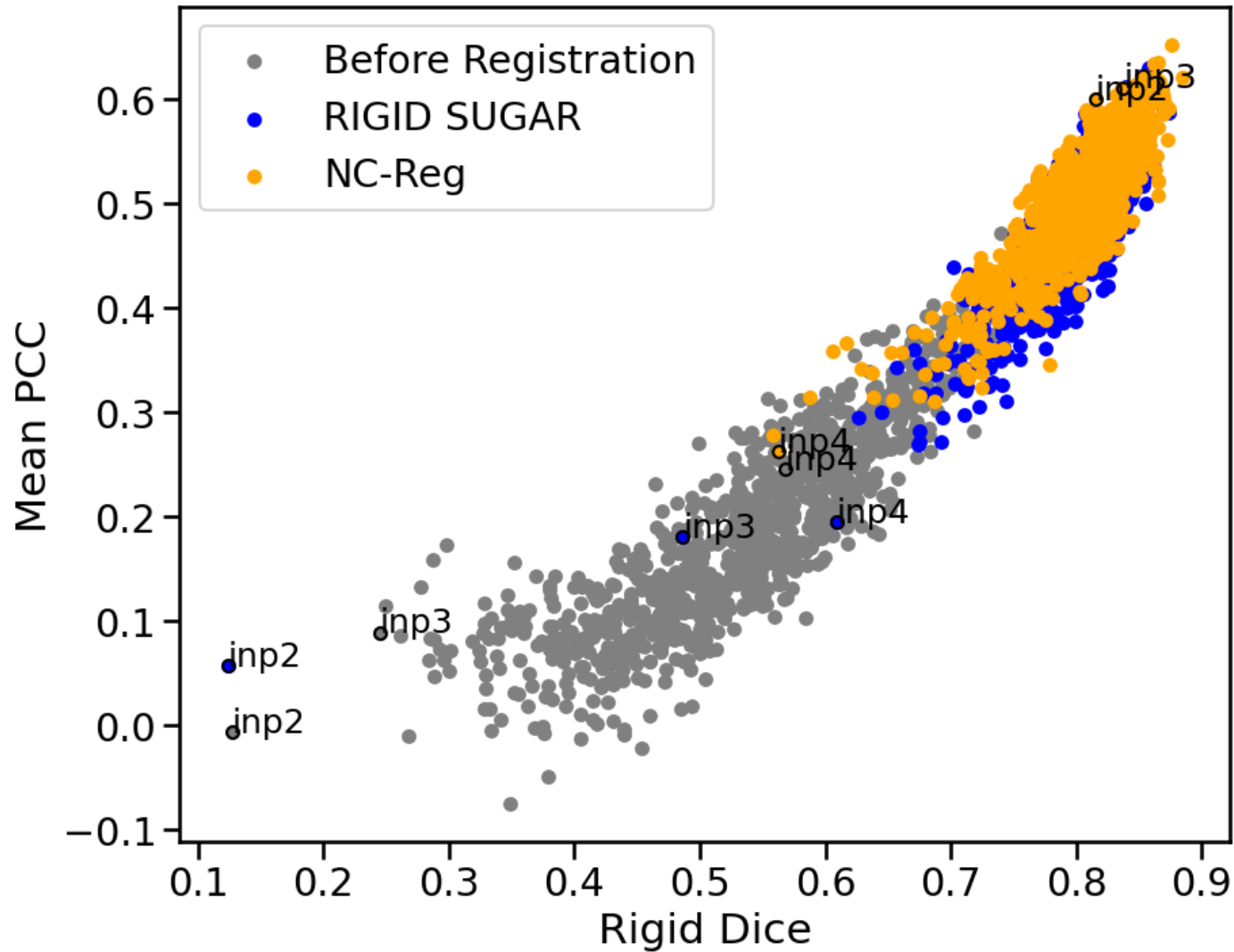


NC-Reg : Integration in NRR workflows

NRR on the *perturbed* dataset

Method	MSE ($\downarrow$)	PCC ($\uparrow$)	Dice ($\uparrow$)	Time (s)	Worst Dice
	sulc	sulc			
SD	0.057	0.912	0.812	35.55	0.0%
NC-Reg + SD	0.056	0.914	0.821	30.26	0.0%
SUGAR	0.167	0.731	0.412	1.46	7.7%
NC-Reg + SUGAR	0.063	0.903	0.822	2.13	0.0%

Key Takeaways



Robust, Consistent and Accurate



Runtimes



Integrate NC maps into an end-to-end neuroimaging pipeline

Thank You !

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April 11, 2026



References

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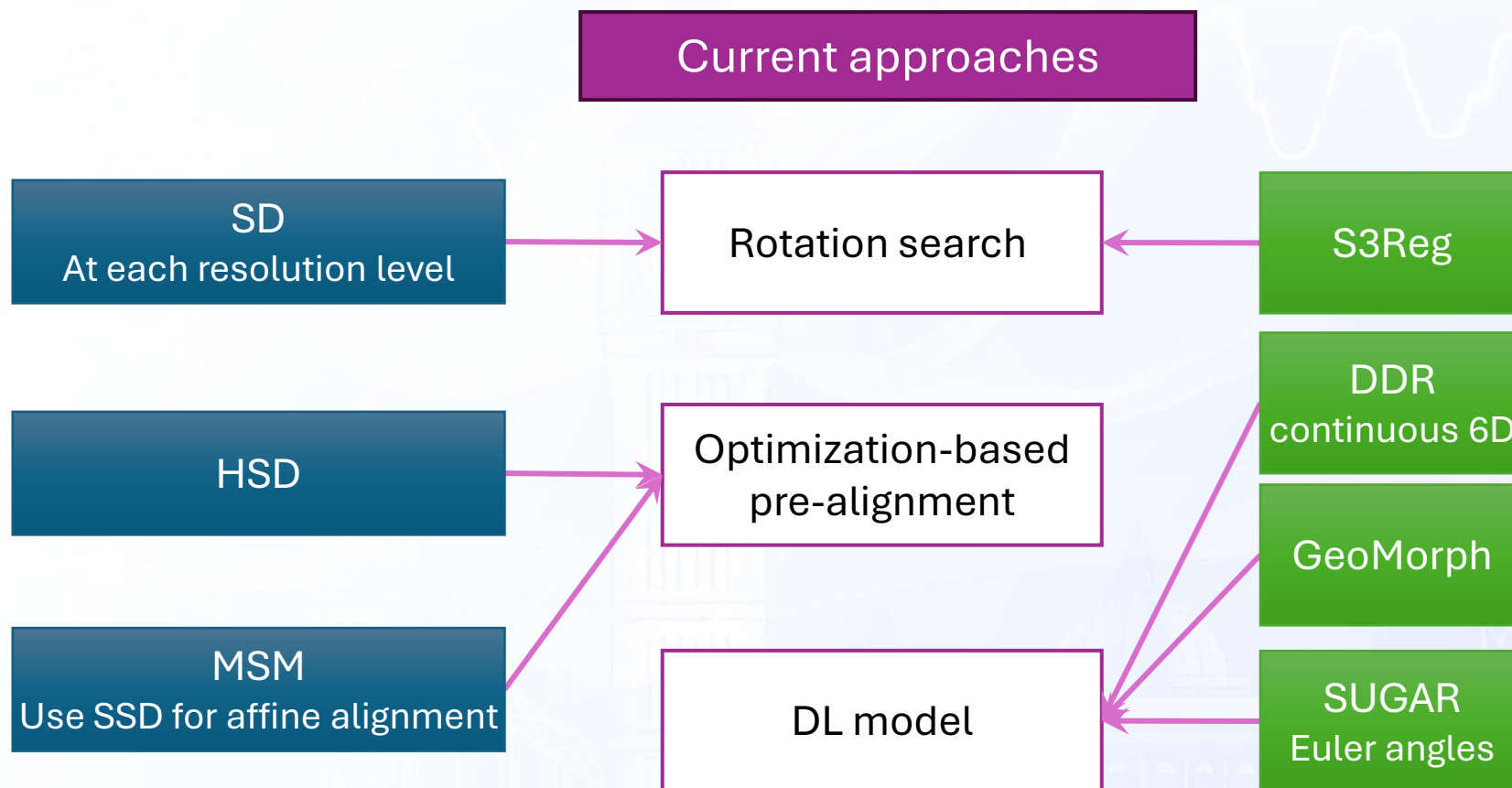
Appendix

April 11, 2026

Ines VATI



Spherical Pre-Alignment



Neural Cortical Maps (NC) - Definition

Model

Encoding + Fully-fused
MLP from [tiny-cuda-nn](#)

$$\mathcal{S}_\varphi: \mathbb{S}^2 \rightarrow \mathbb{R}^{n_f}$$

$$\arg \min_{\varphi} \int_{p \in \mathbb{S}^2} \mathcal{L}(\mathcal{S}_\varphi(p), I(p)) dp$$

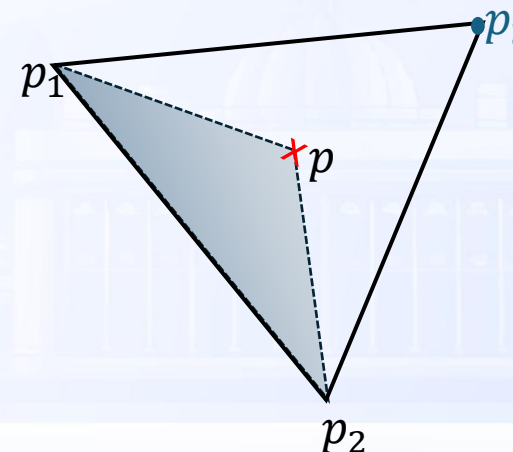
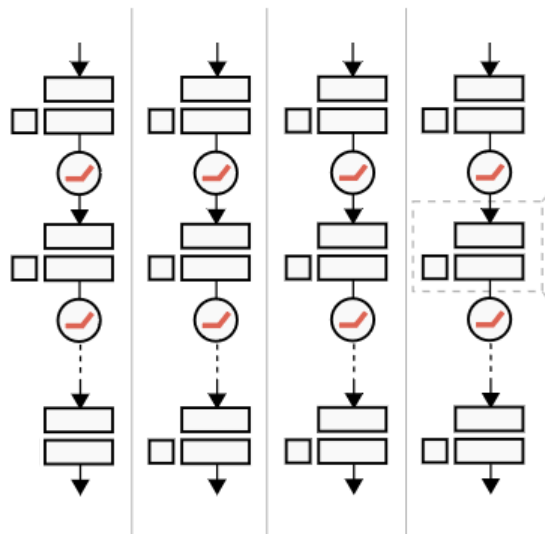
Training

Sampling faces

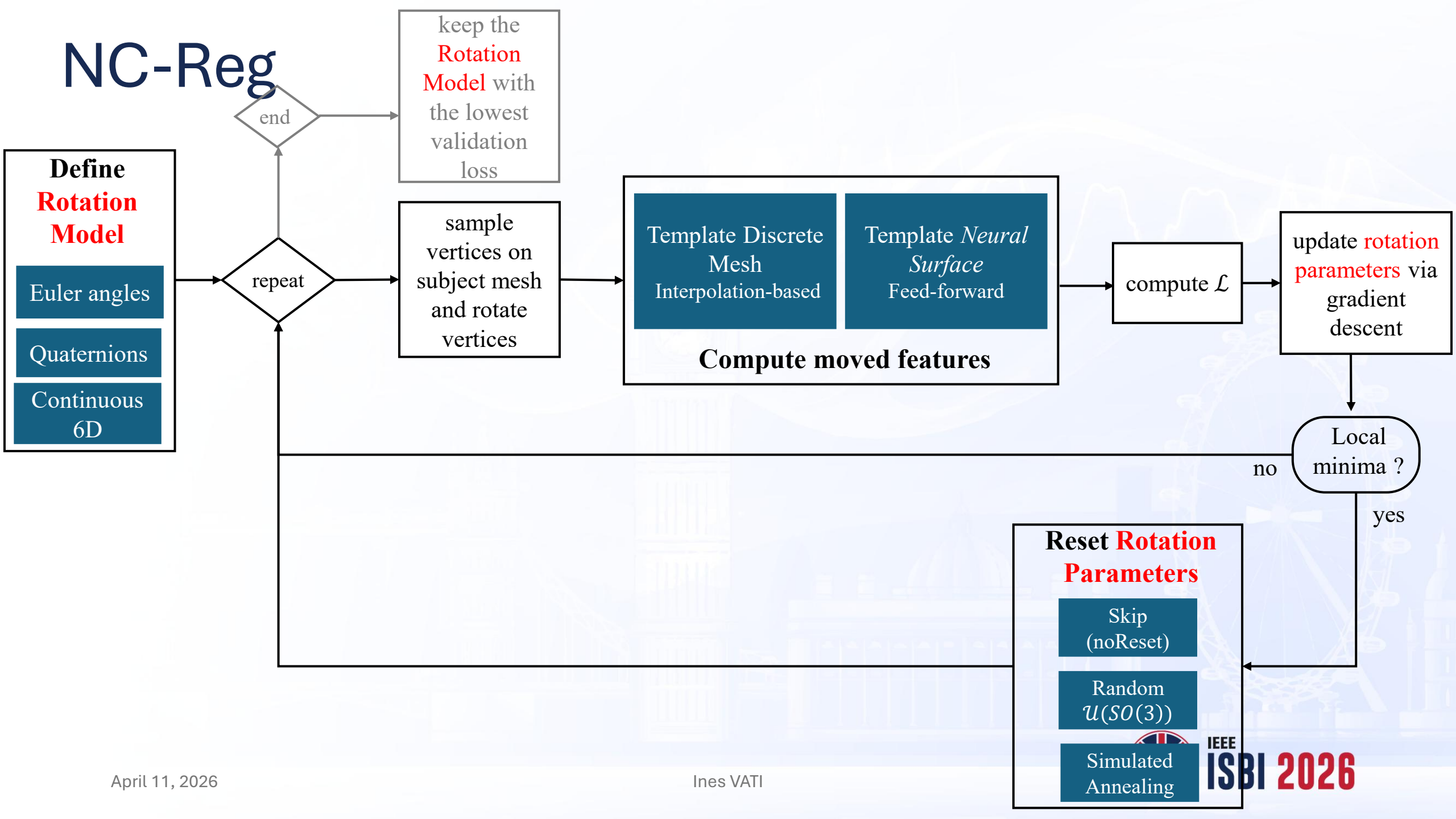
Sampling random
points on the face

Learn interpolation

(Müller et al. 2021)



NC-Reg



Simulated Annealing (SA) reset strategy.

A temperature parameter T_t is decreased every time a local minimum is identified until a temperature minimum $T_{\min}$ is reached. When close to the end of the optimization, the probability of accepting a worse candidate is smaller.

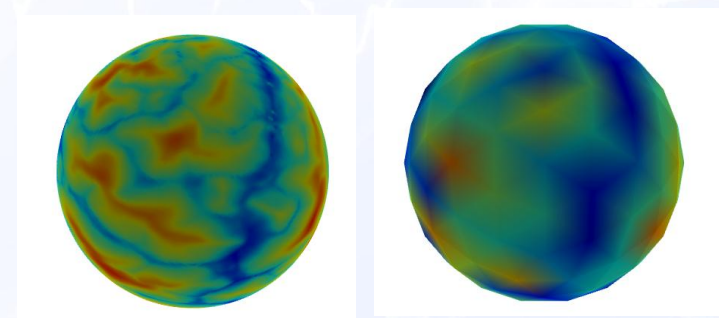
```
1: for  $k \in \llbracket 1, N_{T,\text{iter}} \rrbracket$  do  
2:    $R_{\text{rand}} \sim \mathcal{U}(SO(3))$   
3:    $\Delta\mathcal{L} = \mathcal{L}(\{p_i\}_1^N, F, M, R_{\text{rand}}) - \mathcal{L}_t$   
4:   if  $\Delta\mathcal{L} < 0$  OR  $\left(\Delta\mathcal{L} > 0 \text{ and } \mathcal{U}([0, 1]) < \exp\left(\frac{-\Delta\mathcal{L}}{T_t}\right)\right)$  then  
5:      $R_{t+1} \leftarrow R_{\text{rand}}$   
6:     break  
7:   end if  
8: end for  
9: if  $T_t > T_{\min}$  then  
10:   $T_{t+1} \leftarrow \alpha_T T_t$   
11: end if
```

Neural Cortical (NC) Maps - Memory

Compression for large meshes

Subject file (MB)	# of vertices	136,804
	.sphere	4.57
	.curv	0.51
	.sulc	0.51
	Total	4.59

Reconstruction at any resolution



NC maps	Cartesian inputs Hash # layers: 16 width: 32 neurons # hidden layers: 5	# parameters	(MB)	Median CUDA time (ms)
		955904	3.65	0.18